

MD MOINUL HOSSAIN

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ABOUT

Senior Software Engineer with almost 6 years of experience designing and developing scalable backend applications across industries like Fintech, EdTech, E-commerce, Restaurant Management, and Sales Tax Automation. A fast learner with a proven ability to quickly adapt to new technologies and methodologies, delivering high-quality solutions in dynamic environments. Highly proficient in backend development with expertise in Python, Java, Spring Boot, FastAPI, and cloud platforms like AWS. Skilled in microservices architecture, containerization, and data integration, with extensive experience working with large datasets, complex migrations, and modernizing legacy systems. Demonstrates a strong focus on performance optimization, data integrity, and team leadership, delivering robust and efficient backend solutions for mission-critical systems.

SKILLS

- Programming Languages: *Python, Java, JavaScript, Typescript, C, C++*
- Database: *PostgreSQL, MySQL, MongoDB, Redis, Dynamodb*
- Framework: *FastAPI, Spring Boot, React, Angular*
- Version Control: *Git, GitHub, Bitbucket*
- Containerization: *Docker*
- Cloud: *AWS*
- Technology: *Microservices, Maven, OAuth, CI/CD, Alembic, SQLAlchemy, Pydantic, Kafka, Junit, Spring Cloud, API Gateway, Swagger, Postman, Slack, JIRA*
- Monitoring Tool: *Posthog, Prometheus*
- Cache: *Redis*
- Communication: *English*

WORK EXPERIENCE

Kintsugi AI Inc - US(Remote)

Jan 2024 - Present

Senior Software Engineer

- Spearheaded **sales tax automation processes**, achieving seamless integration across multiple systems for enhanced operational efficiency.
- Managed a dataset of over **129 million records** for address validation, ensuring accuracy and compliance with US regulatory standards.
- Developed **scalable tax rule algorithms** tailored to US-based clients, improving the precision of tax calculations.

- Optimized data storage and retrieval processes using **PostgreSQL**, implementing indexing and partitioning strategies to efficiently handle millions of records.
- Collaborated closely with cross-functional teams to ensure **regulatory compliance**, minimizing errors in tax reporting and enhancing accuracy.

Everforth Co Jp - Japan(Remote)

Aug 2022 - Jan 2024

Senior Software Engineer

- Led a team of 8 developers in the design, development, and deployment of a robust e-commerce platform, ensuring timely delivery of high-quality features.
- Actively participated in project planning, including task assignment, sprint prioritization, and progress tracking, fostering a collaborative and productive team environment.
- Contributed to the development of critical platform features such as checkout, order management, products, inventories and other customer-centric functionalities, enhancing the overall user experience.

Marginedge - US(Remote)

Feb 2021 - Aug 2022

Software Engineer

- Contributed to the development of a comprehensive restaurant management platform that integrated POS systems and automated invoice processing, reducing manual effort for restaurant operators.
- Built and optimized features to streamline real-time cost tracking by automating data flow from sales, labor, and invoices into daily profit and loss statements.
- Implemented robust integration pipelines for over 15 POS systems and major accounting platforms, ensuring seamless data synchronization and error-free transactions.
- Enhanced inventory management functionality by automating product price updates and streamlining inventory tracking for efficiency and accuracy.
- Developed modules for order management, enabling users to place, track, and fulfill orders directly through the platform with real-time updates.
- Ensured secure and efficient handling of financial operations, including bill payments and menu analysis, leveraging cutting-edge technologies to optimize operational workflows.

Emerging IT Bangladesh Ltd - Dhaka, Bangladesh

May 2019 - Jan 2021

Software Engineer

- Developed an edTech system from the ground up, focusing on backend stability and frontend usability.
- Enhanced system performance by 30% through database indexing and caching strategies.

Data Science Trainee

- Conducted statistical analysis and machine learning model training for classification projects.
- Processed and cleaned datasets, preparing insightful reports to guide decision-making.

PROJECTS

EdTech Platform

Technologies: Spring Boot, Docker, Angular, PostgreSQL

- Designed and developed a scalable **educational platform** for students, featuring interactive lessons, quizzes, practice problems, and online exams.
- Conducted **large-scale online exams** with seamless support for **10,000-15,000 concurrent participants**, ensuring near zero downtime and reliable performance.
- Implemented advanced **backend architectures** using **Spring Boot** to handle real-time data synchronization, ensuring a smooth user experience across the platform.
- Designed and maintained a **responsive frontend** using **Angular**, delivering an intuitive interface for students and educators.
- Optimized database performance by structuring efficient schemas, implementing indexing, and fine-tuning queries in **PostgreSQL**, enabling high-speed data access even under heavy loads.
- Utilized **Docker** to containerize and deploy microservices, ensuring consistent builds and simplifying the deployment process across multiple environments.
- Collaborated with a cross-functional team to enhance platform capabilities, contributing to increased engagement and academic success among students.

FinTech Platform Modernization and Migration

Technologies: FastAPI, React, ASP Classic, Visual Basic, Ruby on Rails, MSSQL, PostgreSQL

- Led the migration of a legacy fintech platform from ASP Classic, Visual Basic, and Ruby on Rails to a modern microservices architecture using FastAPI for the backend and React for the frontend.
- Designed and implemented scalable microservices and modular frontend components, ensuring seamless functionality and improved system performance.

- Maintained synchronization and session consistency between the legacy and new systems during the migration process to prevent disruption in financial operations.
- Developed robust database migration strategies to ensure data integrity while transitioning critical financial information to the new system.
- Automated data synchronization workflows to ensure real-time updates and reliability, adhering to stringent fintech compliance standards.
- Gradually phased out the legacy system while ensuring uninterrupted service, reducing downtime and operational risks.
- Worked closely with stakeholders to address security, data accuracy, and performance challenges, achieving a highly reliable and efficient fintech platform.

Sales Tax Automation for US Markets

Technologies: FastAPI, PostgreSQL, AWS, Data Integration, Docker, Opensearch, Posthog, Prometheus, Sentry, Sendgrid

- Designed and developed a **scalable solution** to process and analyze **large datasets** of US addresses, tax rules, and transactions, ensuring compliance with regional tax regulations.
- Integrated multiple external systems to extract **transaction data, customer information, and address details**, enabling comprehensive and automated data pipelines.
- Automated the calculation of **sales tax** for diverse product categories based on complex tax rules and dynamic requirements.
- Leveraged **FastAPI** to develop high-performance APIs capable of handling real-time tax calculations and large-scale data transactions.
- Ensured system **reliability and scalability** by deploying the solution on **AWS**, utilizing cloud services for efficient data processing and storage.

EDUCATION

Bangladesh University of Engineering and Technology - Dhaka, Bangladesh

BSc in Computer Science and Engineering

Feb 2015 - Apr 2019

CERTIFICATIONS & TRAINING

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| ● Neural Networks and Deep Learning | Coursera |
| ● AWS Fundamentals: Going Cloud-Native | Coursera |
| ● Building Scalable Java Microservices with Spring Boot and Spring Cloud | Coursera |

OPENSOURCE CONTRIBUTION

Baledung - Remote

Feb 2022 - Sept 2022

- Contributed in eugenp github repo for Baeldung articles code base

AWARDS & ACHIEVEMENTS

- Gold level for problem solving in Hackerrank
- FRI, Bangladesh Govt, 12 IT Hi-Tech park training on Data Science, ML, AI, Cyber Security, IoT, Blockchain, 2019
- Champion, Hackathon on Blockchain and Web Security, 2019
- 1st Runner-up, Battle of Hardware (IoT), 2019
- Champion, National level Quiz(general Knowledge) Competition
- Participated & won prizes in Math, Informatics, Physics, Chemistry, Biology Olympiads

OTHER CONTRIBUTIONS

Technical Writer

- Contributed in eugenp github repo for Baeldung articles code base

Chat Application

- Worked on backend development of corporate chat application used by largest mobile operators in bangladesh for customer service

Block Programming

- Built block-based visual programming language targeted primarily at children to help learn code.

Smart Home Sweeper

- Built as university project automated robot that sweeps, cleans the floor. For performance improvement this robot can be operated manually also.

Smart Stick for Blind People

- Built as university project for visually impaired people to help them identify any obstacle on their movement